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Integration and optimisation analysis of PLC and SCADA-HMI–IPC systems in intelligent power distribution monitoring

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ABSTRACT

Tahe use of Programmable Logic Controllers (PLCs), Supervisory Control and Data Acquisition (SCADA) systems, and Human-Machine Interface industrial PCs (HMI IPCs) is still limited by the interface latency, inconsistent terminology, and limited operator support. To address these shortcomings in intelligent power distribution monitoring, this research paper presents and justifies a new Integrated Edge-Aware PLC-SCADA-HMI Optimization Model (IEPSHOM) to overcome them. It is novel because it combines a rule-first, decision-tree-refinement step with an adaptive Human-Machine Interface (HMI) to minimize detection latency and enhance the quality of operator responses. The 4-feeder low-voltage substation testbed consisted of Siemens S7-1200 PLCs and adaptive HMI dashboards to compare IEPSHOM with a traditional PLC-SCADA architecture. The outcomes indicate that the accuracy of fault detection in IEPSHOM improved by 11.6 percent, the false-positive rate decreased by 5.5 percent, the average latency decreased by 321 ms to 174 ms, and the operator alert resolution time decreased. There was a 2.4–3.1-point increase in the usability rating, indicating the usefulness of adaptive interfaces in improving decision precision under challenging situations. These results show that the effectiveness of real-time monitoring and operators in digital substations can be enhanced with the help of IEPSHOM.

1. Introduction

Intelligent power distribution monitoring is the coordinated procedure of obtaining, handling, and reacting to data throughout electrical distribution systems with the help of automated digital controls and real-time supervisory instruments [1]. This goal is to make grids more reliable, safe, and energy-efficient by allowing anomalies in the grid to be detected and resolved faster, which is often a part of a grid system incorporating renewable sources, battery storage, and load balancing mechanisms [2]. Here, a Programmable Logic Controller (PLC) is a digital industrial controller that is utilized to provide real-time performance of pre-programmed logic to equipment like circuit breakers, switches, and relays [3]. A Supervisory Control and Data Acquisition (SCADA) system is a remote control that operates the different PLCs and enables the centralized control of distributed electrical equipment via real-time data about the current, voltage, frequency, and breaker status [4]. The Human-Machine Interface Industrial PC (HMI IPC) is an edge computing platform, containing interactive dashboards, on-site diagnostics, and arbitration control logic that can be installed together with the SCADA-PLC infrastructure [5]. Nevertheless, existing PLC-SCADA-HMI implementations in power distribution are seldom able to express a shared optimization goal across field control, supervisory acquisition, and the operator interface. The current research paper contributes to a combined edge-conscious architecture in which detection, visualization, and operator support are in-block designed to minimize latency and reduce interpretation errors during real-time distribution monitoring.

PLCs are field-level devices used in power distribution architectures to provide low-level control and sensor interactions. The SCADA systems operate at the supervisory level, meaning they collect and send information from the distributed PLCs to a central server [6]. The HMI IPCs are increasingly deployed at the intermediate level, enabling visualization of localized areas, real-time analytics, and control decisions based on the edges between field-level hardware and supervisory systems [7]. This three-tier design can enable decentralized power resources to be more autonomous and resilient, especially during dynamic or faulty grid conditions [8]. Although widely adopted across industries, existing practices tend to treat PLC, SCADA, and HMI/IPC devices as discrete systems, with no unified optimization models. For example, Dakheel [6] demonstrated wireless integration among a frequency inverter, a PLC, and an HMI interface, yet the system was tested only in a single-motor lab configuration. Similarly, Waqas and Jamil [4] applied a SCADA-

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HMI interface to monitor hybrid solar-diesel generation, reporting reduced maintenance, but their results were limited to one industrial pipeline site. To situate the proposed framework, Table 1 contrasts recent PLC–SCADA–HMI, edge-enabled SCADA, and intelligent HMI studies with respect to integration level, latency reporting, operator support, and field validation, thereby highlighting the remaining gaps targeted in this work.

Although these studies suggest feasibility, they consistently lack validation under high-speed event loads, cross-vendor interoperability, or embedded optimisation. For instance, Huo et al. [9] showed that PLC signals can support fault detection via time-series forecasting, but this depended on pristine signal quality and did not involve SCADA-HMI coordination. Melo et al. [5] applied IoT-enhanced SCADA logic for predictive maintenance but struggled with communication consistency in signal-limited environments. Studies [10,11] have addressed cybersecurity and digital twin diagnostics, respectively, but did not incorporate unified PLC–SCADA–HMI integration. Kamil et al. [12] proposed a SCADA–PV interface using TIA Portal and PLCs, achieving a voltage response within 250 ms, although the solution remains at the prototype stage. These studies do not resolve three persistent issues in distribution monitoring. First, detection logic is often executed in a single stage, making it sensitive to noise, topology changes, and vendor-specific data formats. Second, SCADA–HMI coupling is typically static, so operator dashboards do not adapt to fault severity, feeder identity, or alarm concurrency. Third, most reports are based on simulated or single-site laboratory conditions, providing limited evidence on end-to-end latency when PLC, SCADA, and an edge HMI must operate concurrently. An architecture that integrates a rule-first screening layer with a learning-based refinement step and exposes its outputs through an adaptive HMI has not been field-validated on a multi-feeder power distribution testbed.

A critical limitation across this body of work is the absence of fully integrated, field-validated architectures. Current studies rarely combine PLCs, SCADA, and HMI/IPC into a synchronized, intelligent monitoring framework. Ngang et al. [13] implemented a low-cost SCADA–PLC system for 33 kV substations. Bouaisha et al. [14] connected rooftop PVs via SCADA–PLC logic. However, neither study introduced real-time edge optimization or dynamic interface adaptation. Rashad et al. [7] reported improved throughput in a petroleum system using SCADA–HMI. Yet this design lacked transferability to power distribution contexts. Recent edge-computing approaches in distribution automation have moved analytical tasks such as anomaly scoring, fast voltage sag detection, or harmonic monitoring to substation-level devices to reduce round-trip times to central SCADA servers [15]. Parallel work on intelligent HMI has shown that context-aware alarm grouping and progressive disclosure of circuit details can lower misinterpretation rates during compound events [16]. In fault detection, shallow tree-based classifiers and boosted ensembles have been used to discriminate between equipment faults, protection malfunctions, and communication artefacts, but these models were not embedded in a PLC–SCADA–HMI pipeline operating under real feeder conditions. The present architecture aligns with these trends but differs in integrating detection and interface adaptation at the edge.

Previous studies often described PLCs, SCADA systems, and HMI/IPC systems as separate technical modules rather than as parts of a unified system. In this paper, they are reframed within an integrated research perspective that emphasizes their combined role in enabling intelligent monitoring. This paper fills two research gaps. It is technically progressive in the field of real-time integration and embedded optimization of PLCs, SCADA systems, and HMI PC platforms. At the operational level, it improves the system interface's usability through adaptive interface design and secure data handling across all automation layers. Despite increasing interest in multi-layer automation, no study has achieved a unified, real-time, and optimized integration of PLC, SCADA, and HMI IPC systems, validated in active power distribution settings, that combines both edge-based intelligence and operator-centered control performance.

The current study seeks to (i) develop and deploy a single PLC-SCADA-HMI IPC architecture to monitor intelligent power distribution, (ii) design embedded optimization algorithms in the IPC edge layer to develop adaptive fault detection, (iii) create a dynamic operator interface that responds to the real-time grid conditions, and (iv) compare the proposed architecture against the traditional ones in terms of detection accuracy, latency, and operator efficiency. The main contributions are as follows: (1) an integrated, edge-aware PLC–SCADA–HMI architecture in which the HMI terminal participates in detection, visualisation, and operator support; (2) a hybrid fault detection pipeline that combines rule-based screening with a shallow decision tree to sustain high accuracy under noisy and heterogeneous operating conditions; (3) an adaptive HMI mechanism that reconfigures dashboards, alerts, and interaction zones according to feeder, event type, and operator workload; and (4) a rigorously instrumented four-feeder testbed evaluation that reports end-to-end timing across PLC, SCADA, and HMI layers and compares the proposed system with rules-only, machine-learning-only, and conventional PLC–SCADA baselines.

2. Related work

Interoperability across PLC, SCADA, and HMI systems is foundational for real-time automation in power distribution. Studies have implemented diverse protocols, such as OPC UA, Modbus, and DNP3, to harmonize multi-vendor devices. Guaillas et al. [17] demonstrated a unified PAC–SCADA–HMI testbed using OPC UA to reduce protocol conflicts in a microgrid environment. Similarly, Pokane et al. [18] proposed a layered IoT–SCADA architecture integrating PLCs, RTUs, and cloud services, which was evaluated using Multi-Criteria Decision Making. The Modbus-DNP3 hybrid model was capable of real-time coordination but relied heavily on the simulation. The performance of OPC UA under real-time constraints was validated by Ladegourdie and Kua [19], who reported sub-50 ms round-trip times and high resilience of client–server communication. Although it was useful, it was not deployed in non-controllable settings. Nogueira et al. [20] evaluated

Table 1
Recent PLC–SCADA–HMI, edge-enabled SCADA, and intelligent HMI studies.

Study	Integration level	Operator / HMI	Validation setting	Performance	Main limitation
Melo et al. [5]	PLC, SCADA, and HMI are coordinated but not jointly optimised	Static dashboards, alarm view	Simulation only	Not reported	No field deployment; no adaptive HMI; no hybrid (rule + ML) detection
Dakheel [6]	Device-level PLC–HMI–inverter, single-line	Basic device status, start/stop	Lab, single-motor setup	Not reported	Not scalable to multi-feeder substations; no fault detection logic
Waqas and Jamil [4]	SCADA–HMI linked; PLC logic mostly independent	Real-time dashboards, maintenance warnings	Single industrial/pipeline site	Not reported	No edge intelligence; no multi-source anomaly discrimination
Ngang et al. [13]	Good SCADA–PLC; no edge HMI / IPC	Supervisory alarms only	Field (33 kV substation)	Not decomposed	No adaptive interface; no hybrid pipeline; limited to status acquisition
Rashad et al. [7]	SCADA–HMI integration; PLC as external endpoint	Clearer process views, fixed screens	Running industrial system	Improved but not layer-specific	Limited transfer to power distribution; no rule–ML fusion
Li et al. [34]	Edge device + SCADA; HMI not co-designed	Indirect (faster alarms), non-adaptive	Mixed lab/field, edge-oriented	Millisecond-level gains at edge	Single-stage ML; no rule-first screening; no adaptive HMI

inductive-coupling narrowband power line communication on overhead medium-voltage feeders and showed that insertion losses and load-dependent current loops can turn the PLC channel into an erasure channel, indicating that field PLC links in distribution grids require adaptive, edge-aware supervisory integration. Borovina et al. [21] modelled error performance for rural 400 V PLC-based smart metering using contagious error distributions and showed that impulsive noise and nonstationary bit error rates must be accounted for when SCADA traffic shares the same medium. These studies underscore the technical feasibility of protocol-level integration but are often conducted in isolation from real-world power grid conditions.

Industrial case studies offer further validation. Ji and Xu [22] benchmarked IPC–PLC–SCADA integration within a manufacturing plant and confirmed latency under 100 ms, yet applicability to power systems was assumed rather than tested. El Zerk et al. [23] validated MATLAB–Simulink integration with live PLC for SCADA HMI applications, achieving a 20 ms control loop cycle time. In oil and gas contexts, Enemosah and Chukwunweike [24] demonstrated edge-enabled SCADA with heterogeneous PLCs and RTUs across remote field sites, though grid-specific dynamics were not evaluated. Folgado et al. [25] conducted a comprehensive review identifying OPC UA as the most robust interoperability layer, though most studies lacked empirical testing in high-voltage substations. Neis et al. [26] introduced the D-SPADES model-driven engineering approach for Energy Management System applications and demonstrated that SCADA-level functions can be automatically generated from high-level models, but the work did not extend model-driven artefacts to adaptive HMI or edge-resident fault detection. The collective gap across these works lies in the absence of system-wide PLC–SCADA–HMI interoperability frameworks tested in operational power distribution environments with heterogeneous vendor platforms.

Real-time optimisation at the edge is critical for fault isolation and system resilience [27]. Liang et al. [28] conducted field studies during extreme weather events, showing that edge-integrated SCADA systems isolated feeder faults within 100 ms faster than central SCADA alerts. Pulakhandam and Nandikonda [29] extended this by consolidating SCADA, HMI, and historian nodes into redundant edge platforms, cutting downtime by 80% and reducing false alarms through localised analytics. Khan et al. [30] implemented a genetic ensemble model using turbine SCADA data, improving detection precision and embedding the model into edge nodes for immediate operator alerting. While these deployments are effective, in many cases, they were aimed at generating assets or at one-time use rather than at distribution grids. According to Baban [31], PLC logic sensor integration was employed to identify vibration anomalies, thereby enhancing detection efficiency by 70%, but there was no scaling of detection, and the multiple-node architecture was not used.

The promise of edge computing is also expressed by Niu et al. 33, who used SCADA feeds locally within fault-detection algorithms and found that detection time was 200 ms, which is 60 times faster than a central system. The reviewed edge-deployed approaches are lightweight AI-based and rule-based, all of which have previously reported latency of less than 300 ms [28,32]. Yet they have observed some endemic problems in implementing such systems with current SCADA and PLC systems. Srinivas et al. [33] proposed a hybrid state forecasting and state estimation paradigm, where Supervisory Control and Data Acquisition and Advanced Metering Infrastructure data were fused using deep neural networks and robust extended Kalman filters to provide observability for accurate distribution measurement, and were not used to control in the adaptive HMI loop. Notably, most implementations lack harmonized fault-handling logic at both the SCADA and HMI interface levels. The main weakness in all these studies is that they have a siloed optimization focus, which does not incorporate adaptive detection into the entire PLC-SCADA-HMI stack with a single control logic and latency metrics benchmarking.

Edge-based fault-detection systems offer high diagnostic speed, but their deployment and domain isolation are inconsistent. Li et al. [34] implemented a cloud-edge collaborative voltage optimisation scheme, which pre-computational reactive-voltage partitions at cloud and activated minute-level correction responses at edge devices were used to demonstrate the viability of distributed optimisation without considering operator-friendly visualisation of multi-feeder faults. Manu [35] employed microcontrollers with wireless sensors, achieving sub-100 ms fault detection and 92% accuracy. However, its focus on low-voltage feeders limited generalisation. Mellit et al. [36] introduced IoT-edge integration for SCADA visualisation in PV systems, reducing manual inspections by 35% but offered limited edge computing load analysis. Yang and Wang [37] deployed ML-based edge detection on industrial rectifiers, yielding a 93.7% detection rate, though it lacked integration with HMI visualisation. Waqas and Jamil [4] used GSM-enabled RTUs for fault alerts within 5 s, suitable for rural contexts but unsuited for high-speed switching networks. Wang et al. [38] designed the YOLO-SS-tiny lightweight detection model with Sigma-CIoU loss and deformable convolution modules to identify substation equipment defects at 274.2 frames per second, but the approach focused on visual inspection and did not bind detections to PLC–SCADA alarm channels or adaptive dashboards.

Iqbal and Baig [39] integrated anomaly detection and time-series forecasting within SCADA, enhanced by edge modules, which improved forecast accuracy and enabled real-time anomaly detection. Still, its application was limited to energy generation. Ferlito et al. [40] developed microservices for edge-local SCADA alerts compliant with IEC standards, but omitted high-voltage fault categories. Simulation-based works [41,42] deployed deep learning at the edge and demonstrated sub-200 ms response time and cloud bandwidth reduction. However, no empirical deployments were shown. Across these contributions, the principal gap remains the absence of fault detection systems that link edge-level analytics with real-time SCADA–HMI visualisation and control logic tailored to power distribution complexity.

Human–machine interface optimisation remains underdeveloped in power monitoring systems, despite its significant impact on operational safety and decision-making. Gupta et al. [43] developed an HMI for microgrid energy scheduling, enabling real-time source switching within 2 s and reducing grid disturbances by 18%. Visual dashboards improved situational awareness by 40%. Yang et al. [44] surveyed advanced HMIs in cyber-physical systems, finding adaptive user interfaces reduced response times by 25–35% and introduced data reprioritisation based on grid volatility. However, neither study benchmarked their HMI performance in large distribution networks. Santillan et al. [45] optimised interface adaptivity via colour-coded PLC fault zones, cutting minor alerts by 80%. Their food-industry context limits direct transfer to substations without validation.

Ojo et al. [46] reviewed interface designs that integrate SCADA and HMI for microgrid control. He showed that real-time operator triggers reduced voltage deviation by 12%, but the findings were model-based. Gladysz et al. [47] introduced adaptive interfaces that reduced cognitive load in substations by 20%, though most results came from pilot sites. Hu et al. [48] tested eye-tracking to optimise HMI panel layouts for critical metric detection, yet did not apply this to live SCADA systems. While these studies support the evolution of HMI, few integrate adaptive dashboards with embedded SCADA-HMI fault logic into a unified operator feedback loop. The significant gap is the lack of adaptive HMI architectures directly linked to fault classification models and edge-triggered alarms in real-time substation control settings.

To address these layered gaps, this study proposes a unified Integrated Edge-Aware PLC–SCADA–HMI Optimisation Model (IEPSHOM) (see Fig. 1). This model consolidates three architectural layers: (i) PLCs as the field control base, (ii) SCADA as the supervisory data coordinator, and (iii) HMI IPC as the edge analytics and visualisation node. The system embeds fault-detection algorithms directly into the HMI IPC, enabling sub-200 ms real-time alerts. The adaptive dashboards are dynamic, adapting to the volatility of the grid and operator feedback.

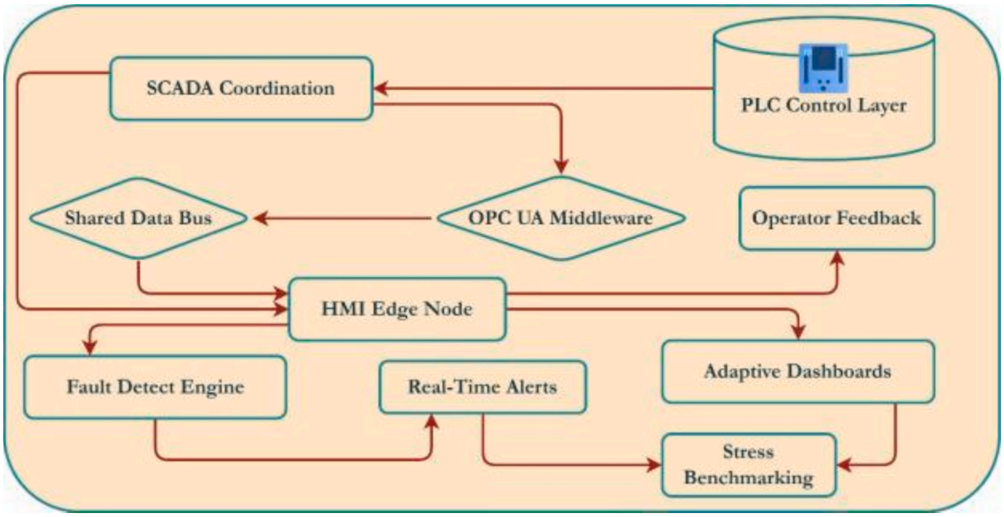

Fig. 1. The architecture of the IEPSHOM, that is, with the control layer of PLC, the SCADA coordination, the OPC UA middleware, the shared data bus, the HMI edge node, the fault detection engine, the real-time alerts, the adaptive dashboards, the operator feedback, and the stress benchmarking.

OPC UA middleware ensures interoperability, and a common semantic data bus is used; performance is determined by load, latency, and operator stress.Fig. 2.Fig. 3..

3. Methods

The experimental platform was assembled with four Siemens S7-1200 PLCs, equipped with digital I/O modules, driven by SiTec PSU6200 supplies, and housed in an IP65-rated enclosure. Advantech UNO-2484G industrial PCs running Intel i7 processors, 16 GB RAM, and solid-state storage were used as hosts for edge processing and supervisory functions, as they can operate without fans in a 10–45 °C temperature range. Electrical variables were captured via calibrated current and voltage transformers, with analog inputs digitized at 16-bit resolution. Control switching employed Siemens SIRIUS contactors, with electromagnetic shielding ensured by ferrite suppression and grounding. Communication and synchronization were achieved using OPC UA with a 100 ms refresh rate, VLAN separation, and GPS-based NTP synchronization (±2 ms). Latency was quantified using three synchronized timestamps: tPLC from the S7-1200 cyclic log, tSCADA from WinCC RT event entries, and tHMI from the InduSoft runtime alert log. All nodes were locked to the TBR-1000 stratum-1 NTP source with drift not exceeding ± 2 ms. PLC to SCADA latency was computed as tSCADA minus tPLC, SCADA to HMI latency as tHMI minus tSCADA, and end-to-end latency as tHMI minus tPLC. Network-induced delay was profiled using a Netropy N61 emulator, which introduced 20–80 ms of additional latency and 1–3% packet drop. Operating-system jitter on the industrial PC was bounded by pinning the HMI runtime to a high-priority process class; measurements exceeding 250 ms were marked as outliers and repeated under the same load. Stress tests under simulated load confirmed reliable operation with latency below 110 ms and minimal jitter.

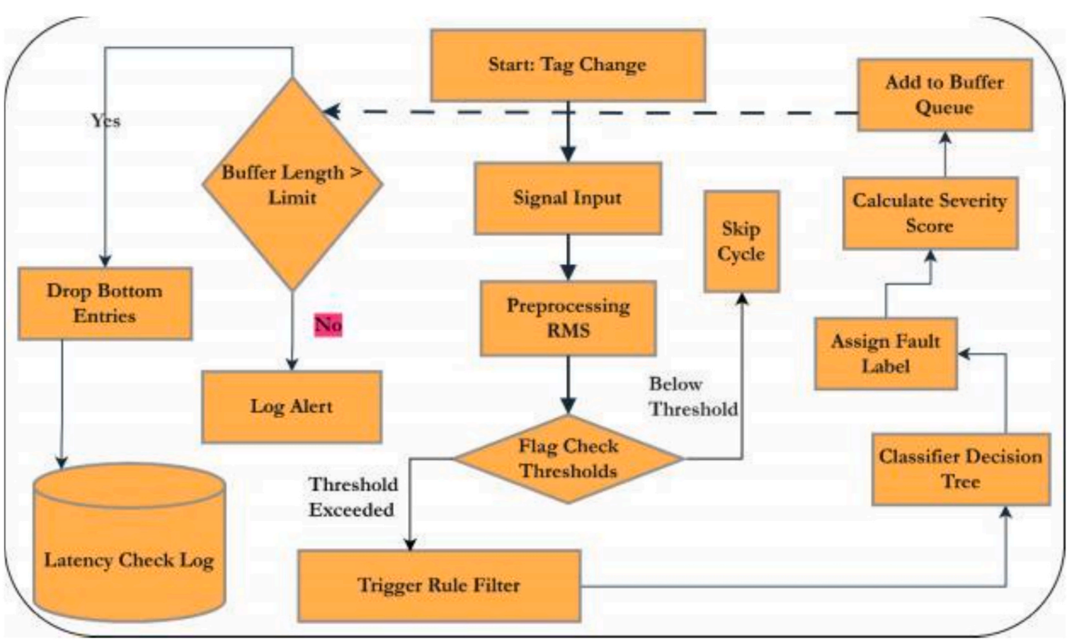

Fig. 2. Fault detection loop with tag change, signal input, RMS preprocessing, threshold check, rule filter, classifier decision tree, fault label assignment, severity score, buffer queue, buffer limit check, drop entries, log alert, skip cycle, and latency check log.

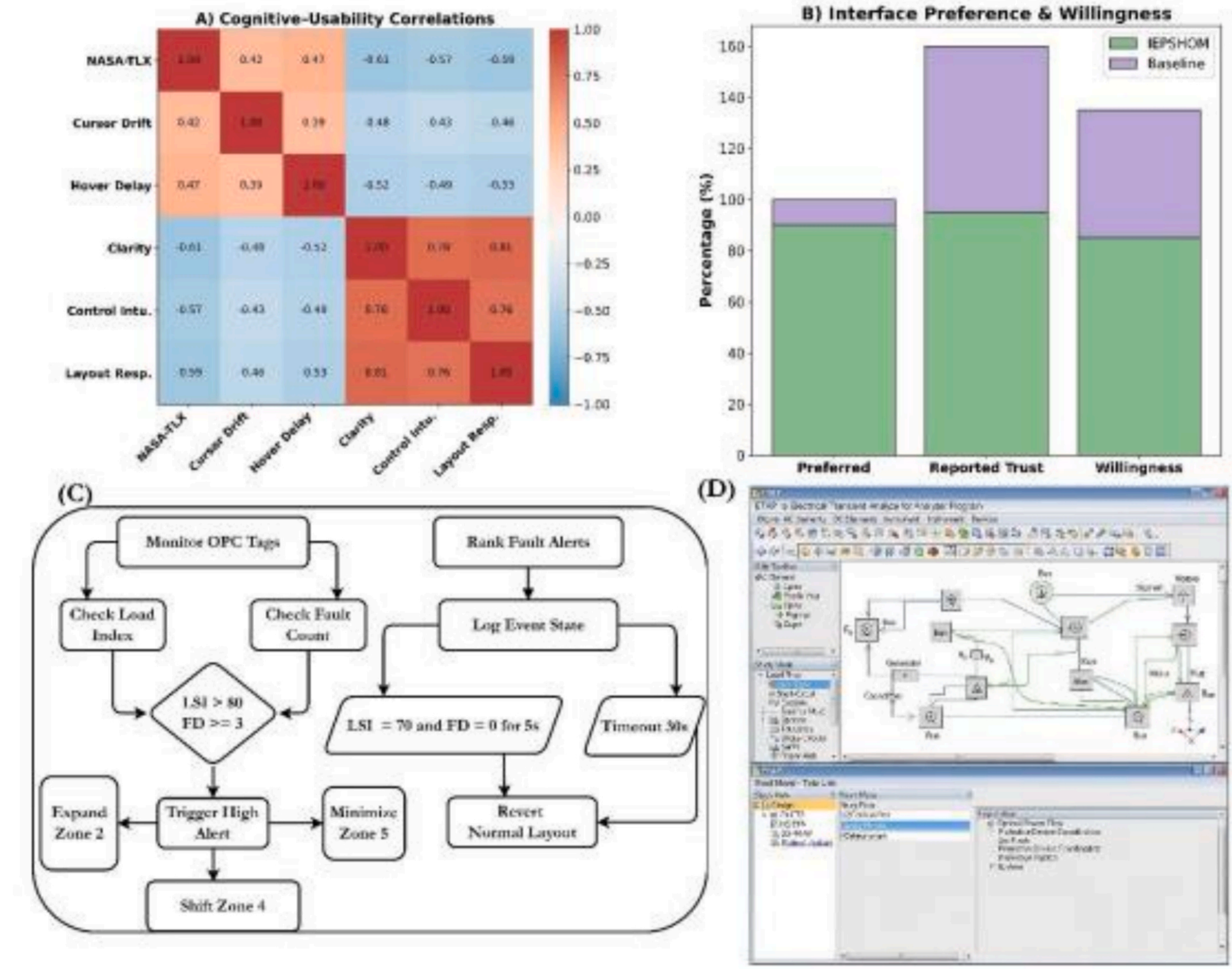

Fig. 3. A) Cognitive usability correlation matrix, B) interface preference, reported trust, and willingness, C) adaptive HMI logic flow, D) ETAP simulation workspace.

4. Edge-Layer optimization Algorithm Development

Three-phase analog signals from Pact RCP-1000 current transformers (*Phoenix* Contact GmbH, Germany) and PTL-110 voltage transformers (Arteche Group, Spain) were sampled at 100 Hz using SM1231 AI modules (Siemens AG, Germany), with 4–20 mA inputs delivered to S7-1200 PLCs (Siemens AG, Germany). Voltage and current values (0–240 V and 0–200 A) were normalized in the PLC memory using calibrated min–max scaling and cyclically written to OPC UA buffers at 100 ms intervals.

The edge-resident analytics modules computed the electrical features used by both the rule and classifier layers according to the following definitions. RMS voltage was computed over a 10-sample sliding window on the IPC (UNO-2484G, Advantech Co., Ltd., Taiwan) using:

$$V_{\mathrm{RMS}}(t) = \sqrt{\frac{1}{N}\sum_{i=1}^{N} V(t_i)^2} \tag{1}$$

In Equation (1), voltage sag was flagged if $V_{\mathrm{RMS}} < 204\,\mathrm{V}$ for two consecutive windows. Phase imbalance was identified via a 64-point FFT per window, calculating pairwise angular deviation:

$$\theta_{ij} = \cos^{-1}\left(\frac{\vec{V}_i \cdot \vec{V}_j}{\|\vec{V}_i\|\|\vec{V}_j\|}\right) \tag{2}$$

In equation (2), if any $\theta_{ij} > 25^{\circ}$ for ≥ 300 ms, imbalance was flagged.

Harmonic distortion was estimated by: $\mathrm{THD} = \sqrt{\sum_{n=2}^{5}\left(\frac{V_n}{V_1}\right)^2} \times 100\%$

(3)

In equation 3, THD above 8.0% over two cycles triggered a harmonic alert. Detection was executed in two consecutive stages within a 100 ms control loop. Stage I applied deterministic rules to incoming OPC UA tags and generated a provisional fault label whenever any of the following held for at least two samples: VRMS $<$ 0.85 $\times$ Vnom, maximum phase-angle deviation θmax $> 25^{\circ}$, or THD $>$ 8.0%. Samples that satisfied none of these conditions were cleared without invoking the classifier. Stage II was invoked only for samples flagged by Stage I or for borderline values (0.80 $\leq$ VRMS/Vnom $\leq$ 0.87 or 22° $\leq$ θmax $\leq$ 25°). Stage II used a depth-5 decision tree (minimum leaf size = 8) trained on ΔV, mean θij, THD, and ΔI to refine the fault type. When Stage I and Stage II produced different labels, the Stage II label was accepted if the associated node probability was $\geq$ 0.65; otherwise, the Stage I label was retained and written to the SCADA alert tag.

The fixed thresholds followed IEC 61000–4-30 Class S guidance for distribution-level voltage quality and were aligned to values observed during feeder-sag emulation on the testbed. Voltage was set to 85% of nominal (187 V for 220 V feeders) to capture sags of 200–300 ms or longer while avoiding nuisance triggers during capacitor switching. The angular threshold of 25° corresponded to the highest stable imbalance recorded during three-phase load asymmetry tests. The THD limit of 8% was selected to flag inverter backfeed and harmonic-rich switching states. A sensitivity sweep was performed on the annotated dataset (n = 1,800 events) by perturbing each threshold by ± 5% and ± 10%. Overall detection accuracy changed by ≤ 1.4 percentage points for voltage and ≤ 1.9 percentage points for THD, confirming robustness to sensor noise and minor calibration drift.

Raw signals were denoised using a Savitzky–Golay filter (order 2, window 7), with timestamps synchronized via TBR-1000 NTP server (Meinberg GmbH, Germany). A two-stage detection model ran on the IPC. Stage I applied hard-rule logic,

$V_{\mathrm{RMS}} < 85\%$, $\theta_{ij} > 25^{\circ}$, or THD $> 8\%$ (4).

In equation 4, stage II invoked a scikit-learn decision tree (depth 5, leaf size 8) using inputs ΔV, mean θ_{ij}, THD, and ΔI. Trained on 1,800 labeled events (70:30 split), it achieved 96.2% accuracy, with inference in 48 ms. Execution followed a 100 ms loop; total pipeline latency stayed < 200 ms. Severity index determined as shown in equation (5),

$$\mathrm{S} = 0.4\Delta\mathrm{V} + 0.3\theta_{\max} + 0.3\mathrm{THD} \tag{5}$$

guided buffer purging when queue exceeded 50 entries, ensuring real-time responsiveness.

5. Adaptive HMI Development

Dashboard Configuration

The operator interface was implemented using a 15.6-inch industrial touchscreen display (FPM-7151 T, Advantech Co., Ltd., Taiwan) physically connected to the DisplayPort of the UNO-2484G IPC (Advantech Co., Ltd., Taiwan), with runtime execution supported by Windows 10 IoT. The dashboard layout was designed using InduSoft Web Studio v8.1 (AVEVA Group plc, United Kingdom), incorporating five functional zones: voltage and current plots (Zone 1), active alert banners (Zone 2), timestamped fault logs (Zone 3), manual override controls (Zone 4), and system status diagnostics (Zone 5). These zones were placed relative to fixed x-y screen coordinates and assigned fixed hierarchical refresh rates based on data criticality. OPC UA subscription mode was used to connect Zones 1 and 5 and to update every 100 ms, whereas the SCADA logs in Zones 2–4 were polled every 500 ms.

The use of color logic with an HSL gradient of a severity index S, where events were labeled green if $S < 15$, yellow if $15 \leq S < 30$, and red if $S \geq 30$. Each critical alert triggered a blinking animation with a 400 ms interval and required an operator to acknowledge it, after which no additional interaction with the interface was allowed. Interactive buttons in Zone 4 followed a press-to-confirm structure with a 1.0 s delay, coupled with a 250 ms debounce to prevent false activations. The real-time tag status of WinCC Advanced RT (Siemens AG, Germany) confirmed state transitions, and the deadlines for action completion were met, as shown by a round-trip confirmation time of 300 ms.

5.1. Real-Time Adaptation Engine.

The HMI adaptation logic was implemented using a rule-based engine in the VBScript runtime of InduSoft Web Studio to continuously monitor the OPC UA-bound system tags and detect state changes. Two input parameters governed interface adaptation: (i) the Load Severity Index (LSI), calculated as shown in equation (6)

$$\mathrm{LSI} = \frac{P_{\mathrm{current}}}{P_{\mathrm{rated}}} \times 100\% \tag{6}$$

(ii) Fault Density (FD) was defined as the count of new fault events occurring within a 2.0 s rolling window. When LSI exceeded 80% or FD was greater than or equal to 3, the interface entered a high-alert mode, expanding Zone 2 to 50% of the screen width and shifting Zone 4 upward to enhance visibility. Zone 5, containing non-critical metrics, was simultaneously minimized and replaced with a scrollable alert stream to reduce visual clutter.

Each adaptive state change triggered a soft transition animation lasting 200 ms, which was logged with a timestamp and the corresponding LSI and FD values. Layout reversion was conditioned upon LSI falling below 70% and FD equaling zero for at least 5 consecutive seconds. A safety override returned the layout to its normal configuration after 30 s, even if minor faults persisted. Within Zone 2, cascading fault alerts were dynamically ranked using a composite priority score defined as

$$\mathrm{Priority} = 0.5 \times \mathrm{THD} + 0.3 \times \Delta V + 0.2 \times \theta_{\max} \tag{7}$$

Equation (7) ensures high-severity conditions were prioritized for operator visibility and action.

5.2. Operator Feedback Loop Integration

The feedback system logged all operator interactions with millisecond-level precision in a locally hosted SQLite database on the IPC, with each entry tagged by UI region, command type, response latency, and SCADA event reference. Operator response latency T_{resp} was computed using the expression $T_{\mathrm{resp}} = T_{\mathrm{ack}} - T_{\mathrm{alert}}$, where T_{alert} denoted the SCADA log time of fault display and T_{ack} indicated the corresponding HMI acknowledgment. The system maintained a rolling buffer of the last 50 response cycles to compute the median response time T_{med} for each operator session.

If real-time latency exceeded $1.4 \times T_{\mathrm{med}}$, the interface automatically elongated button visibility durations by 300 ms and expanded all input controls by 10% to reduce cognitive delay. These UI adaptations continued through the following five cycles of alarm until improved performance was noted. The HMI also monitored sequential alert dismissals. If an operator dismissed three non-critical faults in a row without activating related control objects, the system automatically suppressed low-severity alerts for 60 s. The state of suppression was reinstated when in contact with any critical event.

All operator action logs were matched to the fault classification results in Section 3.2 using shared event indices controlled by WinCC Advanced RT. This two-way tagging facilitated ongoing cross-validation of operator behavior against flawed conditions as detected by models, was used to support real-time interface optimization, dynamic control presentation, and model training by refining the approach.

6. Interoperability

A multi-layer OPC UA implementation, the organization of semantic schemas, and the synchronization of protocols were used to achieve interoperability. On each industrial PC (UNO-2484G, Advantech Co., Ltd., Taiwan), KEPServerEX v6.11 (PTC Inc., United States) served as the OPC UA stack, supporting four active channels, one per PLC (S7-1200, Siemens AG, Germany), using 2048-bit RSA encryption and hourly Secure Channel renewal. Each device connection was authenticated via sideloaded certificates issued by an internal certificate authority. Devices polled at 100 ms intervals with queue depth capped at 1,000 messages. Across all PLCs, 344 active tags were defined: 86 per channel covering analog inputs, digital states, diagnostics, and events. Tags were monitored through heartbeat-enabled subscriptions, triggering alerts upon triple delivery failure. Scalability was benchmarked using MatDeck Network Tag Emulator (Adept Scientific Ltd., United Kingdom) at synthetic tag loads of 25, 50, and 100. At maximum load, latency averaged 108 ms (±11 ms jitter) with zero packet loss.

A semantic data schema based on IEC 61850 defined three top-level classes (Device, Event, Measurement), organized by subsystem (Feeders and Breakers). Tag mapping used a one-to-one alias convention: $Alias_{(i,j)} = Zone_i$: $Device_j$: Parameter, with i = 1–3 and j denoting device IDs. The schema incorporated 344 real and 162 virtual tags spanning HMI adaptation, fault classification, and operator logging. All tags were mapped to canonical types (Float32, Int16, Boolean), and namespace alignment was managed centrally in KEPServerEX. Schema consistency was verified weekly via a Python script checking data types, key uniqueness, and tag integrity.

Multi-protocol synchronization among Modbus TCP/IP, OPC UA, and DNP3 was implemented on a single IPC using the FieldServer Toolkit (Sierra Monitor Corp., United States). All endpoints were time-aligned using a TBR-1000 stratum-1 NTP server (Meinberg Funkuhren GmbH & Co. KG, Germany) polling at 16 s intervals. DNP3 used unsolicited analog reporting with 5% deadband; Modbus tags were polled at 100 ms. Timestamp coherence was ensured by cross-validating PLC logs, SCADA events (WinCC RT, Siemens AG, Germany), and OPC tag updates. Netropy N61 (Apposite Technologies LLC, United States) simulated latency spikes (20–80 ms) and packet drops (1–3%), with synchronization drift held below ± 6 ms. Recovery from protocol desyncs ranged from 140 to 170 ms, depending on the fault class.

7. Validation

7.1. Testbed.

Validation was performed on a custom-built four-feeder low-voltage substation equipped with overcurrent relays (REF601, ABB Group, Switzerland), isolation breakers (3WL, Siemens AG, Germany), and programmable load banks (RLC-5000, Megger Ltd., United Kingdom). Feeder voltage was held at 220 V (±2%), with adjustable loads from 2.4 to 9.6 kW. Electrical parameters were monitored via EM6400NG meters (Schneider Electric, France), interfaced to SM1231 analog inputs (Siemens AG, Germany) using 4–20 mA signals. Faults were triggered by relay misconfiguration, load dump-induced underfrequency (−30% input), and open-neutral events using 250 ms contactor isolation (see Fig. 4A, 4B). All 40 trials per test type were randomized and time-stamped using GPS-synced clocks (TBR-1000, Meinberg GmbH, Germany). Each configuration (baseline PLC–SCADA, rules-only, ML-only, anomaly-detection, and IEPSHOM) was executed over 40 events × 3 repetitions, giving 120 runs per configuration and 600 labelled sequences in total. Trials were block-randomised to balance feeder identity and load level, and were performed in a single session to keep ambient temperature, network utilisation, and supply voltage constant. Operator control access was physically restricted.

7.2. Baseline System and Performance.

The baseline system consisted of Siemens S7-1200 PLCs, configured with fixed ladder logic, for the deterministic control of feeder relays and contactors. Supervisory monitoring was executed through WinCC RT Professional (Siemens AG, Germany), which provided centralized data visualization but lacked local edge analytics. The operator interface was delivered on a passive industrial HMI panel (TPC-1251 T-E3AE, Advantech Co., Ltd., Taiwan) without adaptive dashboards, cognitive feedback loops, or embedded classification models. Communication between PLCs and the SCADA runtime occurred through direct cyclic polling at 250 ms intervals, with no OPC UA middleware or semantic schema mapping.

The fault detection was based on pre-programmed threshold triggers in ladder logic only and all alerts were sent to the SCADA layer and without any prioritization or real-time adjustment. This architecture is a traditional PLC-SCADA-HMI system that is commonly used in a small-scale substation, and it is the reference point of assessing performance gains that the IEPSHOM architecture would provide. Three comparative baselines were done on the same four-feeder platform to separate the contribution of each of the detection components. To isolate the contribution of each detection component, three comparative baselines were implemented on the same four-feeder platform. The rules-only baseline executed the Stage I thresholds (voltage 85%, θmax 25°, THD 8%) directly on the PLC and pushed alerts to SCADA without classifier refinement. The ML-only baseline bypassed Stage I and applied the depth-5 decision tree to every 100 ms sample, using forward fill for missing features over a 300 ms window. The anomaly-detection baseline deployed an isolation forest (contamination = 0.05, 200 estimators) trained on 1,200 normal-operation samples and run on the IPC. All baselines used identical OPC UA tag sets, NTP synchronization, and HMI logging. Detection accuracy, false-positive rate, PLC-HMI latency and operator response time were all measured so that they could be fairly compared on a component level.

An UNO-2484G IPC (Advantech Co., Ltd., Taiwan), adaptable HMI, embedded fault classification and protocol bridging was added to the IEPSHOM. The settings were aligned with each system experiencing 40 randomized fault occurrences being repeated three times (120 trials per system) with the loads being adjusted to ± 5%. Trial order was based on block-randomization. The detection accuracy, false positives, response time, and event-to-resolution of normalized system views were independently reported by an external logger (Chronosync NDL-1000, NetBurner Inc., United States).

7.3. Task Simulation

Evaluation of the operators was done with 20 participants randomly

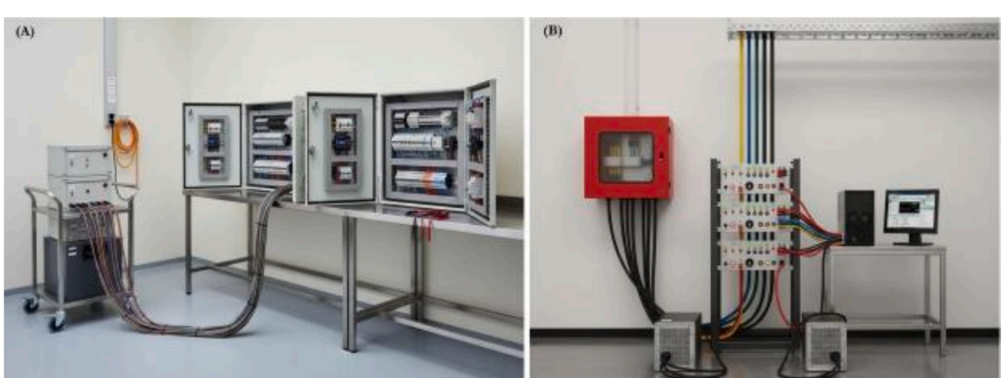

Fig. 4. A) Four-feeder distribution testbed showing the mobile controller cart, multi-core harness, and three open protection/PLC cabinets on a stainless-steel bench for inspection of internal wiring; and B) Edge-enabled PLC–SCADA–HMI configuration showing color-coded feeder inputs from a wall-mounted junction box, front-access measurement and injection panel, two programmable load units, and an industrial PC with HMI runtime for real-time monitoring.

assigned to IEPSHOM or controls. All were subjected to a 20-minute training session on the variants of interface controls, alerts and validation of responses. Sessions were held individually in a controlled ambience of silence. A set of 12 tasks which were accomplished by operators included: (i) fault tagging, (ii) manual redirection of power and (iii) supervisory escalation. There was a 50% increment in task complexity with concurrently faulting feeders. The InduSoft logger was used to write off all interactions in sync with the time of SCADA. Results of the response were classified as correct, delayed, or incorrect and time was noted on completion of the tasks at 10 ms resolution indicated.

7.4. Cognitive Load.

The cognitive load was measured using a passive data recording and self-subjective measures. Keystroke-patterns and hover-delays, control dwell times were logged to obtain attention strain, and a high load was considered to be when alert-to-cursor becomes more than 1.4 times than the median one during the session. NASA-TLX surveys were also administered to participants at the conclusion of each block of a task where they were asked to rate their mental demand, effort, performance, and time pressure. The cursor jitter and the tremor amplitude were recorded through a 60 Hz touchscreen driver as biometric proxies. Changes in the cursor beyond 3.0 mm during alerts showed that there was high stress. All values were set forth by activity and fault discrimination, afterwards incorporated into standardized cognitive load symbol per participant and state.

7.5. Usability.

The Usability testing used a within-subject counterbalanced design, which subjected each subject to the static and adaptive HMI layouts; each subject was allowed to subscribe to both layouts in alternating blocks of tasks. Classifier outputs were used to test performance on an interaction log of the HMI runtime and the SCADA system. Some important measures of usability include time to complete a task, the number of actions required to switch between zones, the rate of misinterpretation of an alert, and a post-task rating on a 10-point Likert scale. The movement of the cursor in and out of the interface zones was termed interface switching, and the misinterpretation was logged when critical alerts were not spotted and lower-priority alerts were processed. ISO 9241–11 questionnaires were used to gather clarity ratings for both configurations.

7.6. Statistical analysis

All analyses were conducted using SPSS Statistics version 29.0 (IBM Corporation, United States). The descriptive outputs were calculated for every system configuration, including detection latency, accuracy, operator response time, and cognitive load index. Independent-samples t-tests were used to assess differences between groups for each of the IEPSHOM and baseline configurations, including latency, false-positive rates, and operator performance scores. A repeated-measures analysis of variance was used to compare the within-subjects static and adaptive HMI interfaces. Usability differences, including interface clarity and task completion times, were tested across interface conditions. All reported significance values were based on two-tailed tests, with a threshold of $p < 0.05$.

8. Results

All four instrumented feeder branches reported highly stable baseline parameters (see Table 2). The nominal voltage ranged from 219.8 ± 2.1 V in Branch Gamma to 220.2 ± 1.9 V in Branch Beta. Mean current varied between 9.4 ± 0.2 A and 9.6 ± 0.3 A across the branches. Active power consumption was consistent, averaging 3.5 ± 0.1 kW to 3.6 ± 0.1 kW. Frequency measurements remained tightly clustered from 49.96 ± 0.02 Hz to 49.99 ± 0.01 Hz. Load levels ranged from 47.9 ± 1.4% in Branch Delta to 50.4 ± 1.5% in Branch Gamma.

Table 2
Baseline operating conditions by instrumented feeder branch.

Protection chain ID	Nominal voltage (V)	Mean current (A)	Active power (kW)	Frequency (Hz)	Load level (%)
REF601–3WL–RLC-5000 (Branch Alpha)	220.1 ± 1.8	9.4 ± 0.3	3.5 ± 0.1	49.98 ± 0.02	48.5 ± 1.3
REF601–3WL–RLC-5000 (Branch Beta)	220.2 ± 1.9	9.5 ± 0.2	3.6 ± 0.1	49.97 ± 0.03	49.0 ± 1.1
REF601–3WL–RLC-5000 (Branch Gamma)	219.8 ± 2.1	9.6 ± 0.3	3.6 ± 0.1	49.99 ± 0.01	50.4 ± 1.5
REF601–3WL–RLC-5000 (Branch Delta)	220.0 ± 1.7	9.4 ± 0.2	3.5 ± 0.1	49.96 ± 0.02	47.9 ± 1.4

Detection performance and latency varied significantly between the baseline and IEPSHOM configurations (see Table 3). The IEPSHOM achieved a detection accuracy of 96.2%, compared to 84.6% for the baseline PLC + SCADA. False positives decreased from 7.9% to 2.4%, and false negatives declined from 7.5% to 1.4%. Mean latency was reduced from 321 ± 48 ms under the baseline setup to 174 ± 31 ms with IEPSHOM. Maximum latency improved from 410 ms to 196 ms under the full system configuration. Because all five configurations were executed on the same four-feeder platform, with identical OPC UA tag sets, NTP synchronization, and randomized fault order, the improvements reported in Tables 3 and 5 reflect architectural differences rather than changes in test conditions. In this aligned setting, only the IEPSHOM configuration met both targets simultaneously: accuracy above 95% and mean end-to-end latency below 200 ms.

Per-class behaviour was examined for the three additional baselines. The rules-only configuration retained high recall for line-to-line short faults at 97.2% with a precision of 96.0%, but its performance dropped on open-neutral faults, where precision fell to 91.4% and recall to 93.7%. The ML-only configuration balanced the two metrics across all classes, with precision ranging from 95.2% to 97.5% and recall ranging from 96.4% to 98.1%. However, it generated slightly more false alarms on underfrequency sags than IEPSHOM. The anomaly-detection configuration achieved acceptable recall (94.6% for line-to-line, 93.5% for underfrequency, and 93.1% for open-neutral) but remained below the hybrid pipeline on open-neutral discrimination. By contrast, IEPSHOM maintained the strongest overall profile, with 97.4% precision and 99.1% recall for line-to-line faults, 98.3%/98.3% for underfrequency sags, and 96.6% precision with 97.4% recall for open-neutral faults.

Operator responsiveness and alert resolution efficiency differed across configurations (see Fig. 5A). The IEPSHOM recorded a median response time of 1013 ± 122 ms, compared to 1448 ± 210 ms under

Table 3
Fault detection accuracy and latency across system configurations.

Configuration	Detection accuracy (%)	False positives (%)	False negatives (%)	Mean latency (ms)	Max latency (ms)
Baseline PLC + SCADA	84.6	7.9	7.5	321 ± 48	410
Rules-only	90.3	5.6	4.1	225 ± 32	338
ML-only	94.9	3.7	3.0	188 ± 29	252
Anomaly-detection	92.7	4.9	3.8	210 ± 34	289
IEPSHOM	96.2	2.4	1.4	174 ± 31	196

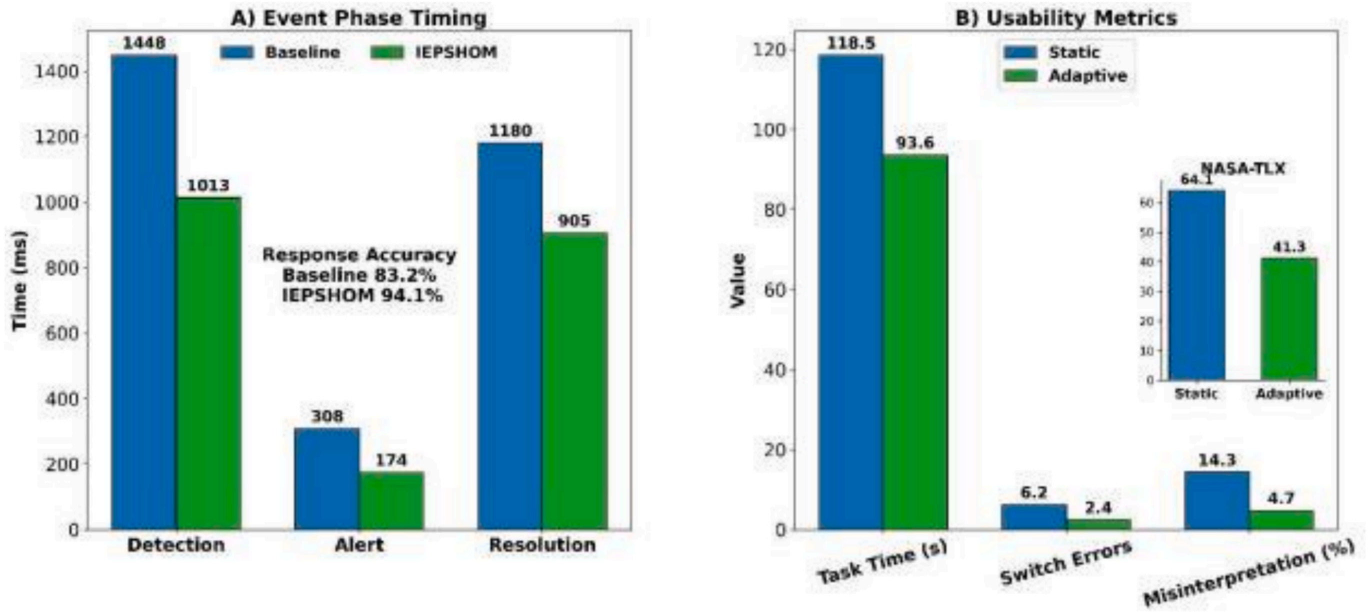

Fig. 5. A) Event phase timings for detection, alert, and resolution with response accuracy for Baseline and IEPSHOM. B) Usability comparison of static and adaptive HMI showing task time, switch errors, misinterpretation rate, and NASA-TLX cognitive load.

the baseline PLC + SCADA. Event-to-alert delay was reduced from 308 ± 36 ms to 174 ± 27 ms. The alert-to-resolution time decreased from 1180 ± 95 ms to 905 ± 76 ms. Response accuracy increased from the baseline setup (83.2%) to the IEPSHOM setup (94.1%).

The layout differences on performance were indicated by interface usability and cognitive load measures (see Fig. 5B). The time and duration of completion of tasks were reduced by 24.9 s from 118.5 ± 11.4 s in the baseline HMI to 93.6 ± 9.2 s in the adaptive IEPSHOM HMI. The misinterpretation rate of alerts decreased to 4.7%. The average NASA-TLX cognitive load score decreased from 64.1 ± 6.7 to 41.3 ± 5.9, indicating greater operator efficiency with the adaptive interface design.

Fault detection and classification metrics across three fault types demonstrated consistently high model performance (see Table 4). For line-to-line short faults, 114 true positives were recorded, with 3 false positives and 1 false negative, yielding a precision of 97.4%, a recall of 99.1%, and an F1-score of 98.2%. Underfrequency sag faults had 117 true positives, 2 false positives, and 2 false negatives, with precision, recall, and F1-score of 98.3%. Open-neutral faults resulted in a 97.0% F1-score from 113 true positives, 4 false positives, and 3 false negatives.

Timing metrics indicated substantial latency reductions under the IEPSHOM (see Table 5). The mean event-to-alert delay was 308 ± 36 ms in the baseline PLC + SCADA system, compared to 174 ± 27 ms under IEPSHOM. Median operator response time improved from 1448 ± 210 ms to 1013 ± 122 ms, while alert-to-resolution duration decreased from 1180 ± 95 ms to 905 ± 76 ms. The 90th percentile of total latency dropped from 1692 ms to 1174 ms. The percentage of events resolved under 1500 ms increased from 58.3% to 91.7%. All configurations were run over 40 events × 3 repetitions (n = 120 per configuration). Across repetitions, the coefficient of variation for event-to-alert delay stayed below 10% for IEPSHOM (CV = 9.4%) and below 15% for ML-only (CV = 13.1%), indicating stable timing behavior. Detection accuracy 95% confidence intervals were: baseline PLC + SCADA 81.0–88.1%; rules-only 87.6–92.9%; anomaly-detection 90.1–94.9%; ML-only 93.1–96.4%; IEPSHOM 95.0–97.4%.

Table 5
Event-to-response timing metrics across system configurations.

Metric	Baseline PLC + SCADA	Rules-only	ML-only	Anomaly-detection	IEPSHOM system
Mean event-to-alert delay (ms)	308 ± 36	225 ± 32	188 ± 29	210 ± 34	174 ± 27
Median operator response time (ms)	1448 ± 210	1185 ± 150	1108 ± 135	1236 ± 165	1013 ± 122
Alert-to-resolution duration (ms)	1180 ± 95	1015 ± 82	958 ± 79	1068 ± 88	905 ± 76
90th percentile total latency (ms)	1692	1385	1242	1458	1174
Percentage resolved < 1500 ms (%)	58.3	79.2	87.5	74.1	91.7

Global SHAP analysis shows RMS voltage as the primary driver (41.3%), followed by harmonic distortion (28.7%), phase imbalance (19.6%), and current variation (10.4%) (Table 6). The high positive-SHAP share for ΔV and THD (>88%) indicates that voltage sags and

Table 4
Fault detection and classification performance across fault types.

Fault type	True positives (n)	False positives (n)	False negatives (n)	Precision (%)	Recall (%)	F1-score (%)
Line-to-line short (L–L)	114	3	1	97.4	99.1	98.2
Underfrequency sag	117	2	2	98.3	98.3	98.3
Open-neutral fault	113	4	3	96.6	97.4	97.0
Total (all faults)	**344**	**9**	**6**	**97.5**	**98.3**	**97.9**

Table 6
Global SHAP feature importance for the decision tree (depth = 5).

Feature	Mean |SHAP| (AU)	Normalized importance (%)	Rank	% positive SHAP
RMS Voltage (ΔV)	0.237	41.3	1	88.2
Harmonic Distortion (THD)	0.165	28.7	2	90.7
Phase Imbalance (θ deviation)	0.112	19.6	3	84.5
Current Variation (ΔI)	0.060	10.4	4	61.9

elevated harmonics typically push predictions toward the fault class, aligning with reductions in false positives when these indicators normalize. Phase-angle deviation contributes meaningfully but secondarily, while ΔI mainly refines borderline cases, consistent with its lowest mean |SHAP| and rank.

Per-fault-type SHAP profiles show ΔV dominates line-to-line shorts (44.8%) and underfrequency sags (48.2%), whereas THD is most influential for open-neutral faults (36.7%) (Table 7). Phase-angle deviation shows consistent differences across categories (∼18–21%), supporting discrimination when ΔV or THD alone is ambiguous. Current variation remains a minor factor (≈9–11%), suggesting limited incremental value beyond voltage and harmonic features. These patterns validate the hybrid rules-plus-tree design: voltage thresholds and harmonic cues trigger faults, while angular imbalance refines boundary decisions.

Task-completion metrics revealed consistent gains for adaptive interfaces across both fault-complexity conditions (see Fig. 6A). For low-complexity single-event faults, mean task completion time decreased from 62.4 ± 4.1 s (baseline HMI) to 48.7 ± 3.6 s (adaptive HMI), with missed acknowledgments reduced from 5 to 1 and interface switches from 211 to 103. Under high-complexity concurrent-event conditions, mean completion time dropped from 87.6 ± 6.2 s to 66.1 ± 5.3 s, missed acknowledgments from 13 to 4, and interface switches from 288 to 145.

Between-group analysis demonstrated significantly improved precision and alert handling in the IEPSHOM group compared with baseline (see Fig. 6B). Correct classification rate rose from 84.1 ± 3.6% to 94.8 ± 2.3% (mean difference = +10.7, $t(18) = 7.82$, $p < 0.001$). Alert prioritization accuracy increased from 78.3 ± 4.1% to 91.6 ± 2.7% (+13.3, $t(18) = 7.19$, $p < 0.001$), while non-critical alert suppression improved from 68.2 ± 5.5% to 92.4 ± 3.1% (+24.2, $t(18) = 10.83$, $p < 0.001$). Redundant actions dropped by 6.8% ($t(18) = -10.02$, $p < 0.001$), and missed critical escalations were eliminated (from 6 to 0).

Measures of user evaluation indicated higher clarity and satisfaction with the IEPSHOM interface than with the baseline system (see Fig. 7A). Clarity ratings improved from 6.3 ± 0.9 to 8.7 ± 0.6, with a mean difference of + 2.4 ($t(19) = 8.92$, $p < 0.001$). Control intuitiveness ratings increased from 6.1 ± 1.2 to 8.5 ± 0.8 (mean difference = +2.4, $t(19) = 6.77$, $p < 0.001$), while layout responsiveness rose from 5.8 ± 1.1 to 8.9 ± 0.5, indicating a + 3.1 improvement ($t(19) = 10.23$, $p < 0.001$).

The efficiency measures during the completion of the tasks indicated significant decreases in the rates of the users' errors with the IEPSHOM interface (see Fig. 7B). Zone switching actions per task decreased from 8.6 ± 1.2 to 5.2 ± 0.7 (mean difference = –3.4, $t(19) = -8.36$, $p < 0.001$). Input misfires per 10 actions were reduced from 1.1 ± 0.5 to 0.3 ± 0.2 (mean difference = –0.8, $t(19) = -6.92$, $p < 0.001$). Alert dismissal errors declined from 7.9 ± 2.0 to 2.8 ± 1.1, yielding a –5.1 difference ($t(19) = -7.64$, $p < 0.001$).

Table 7
Per-fault-type normalized mean |SHAP| (%) for the decision tree.

Feature	Line-to-Line Short	Underfrequency Sag	Open-Neutral
RMS Voltage (ΔV)	44.8	48.2	31.0
Harmonic Distortion (THD)	26.5	22.9	36.7
Phase Imbalance (θ deviation)	20.1	18.0	21.2
Current Variation (ΔI)	8.6	10.9	11.1

There were strong negative relationships between NASA-TLX cognitive load scores and the three dimensions of usability. Clarity rating was found to have a negative correlation with NASA-TLX of −0.61 ($p = 0.004$), control intuitiveness of −0.57 ($p = 0.008$) and layout responsiveness of −0.59 ($p = 0.006$). Besides, reduced clarity was also associated with the increased cursor drift ($r = -0.48$, $p = 0.034$), whereas the longer hover delay was also correlated with the lack of layout responsiveness ($r = -0.53$, $p = 0.018$), reflecting the inefficiency in the perceptual processes between various load conditions. Higher interface preference, trust and reuse intention were found in the evaluation by the end of the session with the IEPSHOM. The preferred interface was 90.0 percent in The IEPSHOM in comparison to 10.0 percent in the line of baseline ($\chi^2 = 12.8$, df = 1, $p < 0.001$). Trust in functionality of the interface was reported at 95.0 v. 65.0 ($\chi^2 = 6.12$, $p = 0.013$), and willingness to reuse was 85.0 v. 50.0 ($\chi^2 = 5.00$, $p = 0.025$) with IEPSHOM reporting higher trust levels in interface functionality and willingness to reuse interface levels.

9. Discussion

The results of the performance of the IEPSHOM attained the same gains within the baseline basing on the precision of the detection, latency and performance of the operators. The false positive was reduced to 2.4 percent and the fault detection to 96.2 percent and the average event to alert latency also reduced by a half to 174 ms. These changes gave way to higher speed and accuracy of response and lessened the time of alert-to-resolution supported by higher clarity, control and layout scores. These large negative correlations between NASA-TLX and usability indices indicate that the gain in efficiency was not related to accelerated telemetry but to adaptive presentation. The adaptive interface was discovered to be significantly quicker, minimized the misinterpretation index as well as maximized the correctness of user choice taking judgment in complex circumstances. The operator was also more correct and critical alerts were given more priority as well as faster reaction. The optimized interface was always significantly positively correlated with the subjective ratings of the clarity, and the control intuitiveness of the layout responsiveness and strongly negatively correlated with cognitive load. Post-task feedback provided by the system after a system testing established the reliability of system and design preference which under various circumstances sustained the preference of embedded edge analytics, adaptive visual feedback, and semantic interoperability in offering the scalability and the human friendly substation monitoring.

The improvements in the performance of the IEPSHOM (that is, the accuracy of fault detection increases to 96.2% and the shortening of alert latency to 174 ms) confirm cohesively reduced edge analytics in the efficiency of control loops. Guaillas et al. [17] also found that OPC-UA cross-Modbus, PROFINET, and CAN bridging provided cross-protocol situation coherence with sub-50 ms polling and visual feedback update regarding cross-polling typically in 45 ms. Similarly, [27,28] isolated faults in the range of 100 ms and a drop in unnecessary operator requests by one-fourth when edge-sensing relays were used at the time of live storms. Their systems were based on relays intelligence but our findings indicate that a direct implementation of hybrid classifiers, as implemented in the IPC layer, provides better performance when compared to relay-only implementation because it is capable of performing multi-parameter analysis (voltage dips, phase angle shifts, and harmonic distortion) of a 200 ms rolling window. It can be synchronized to identify signal patterns across feeders to reduce false positives and remove response delays because of upstream SCADA processing bottlenecks.

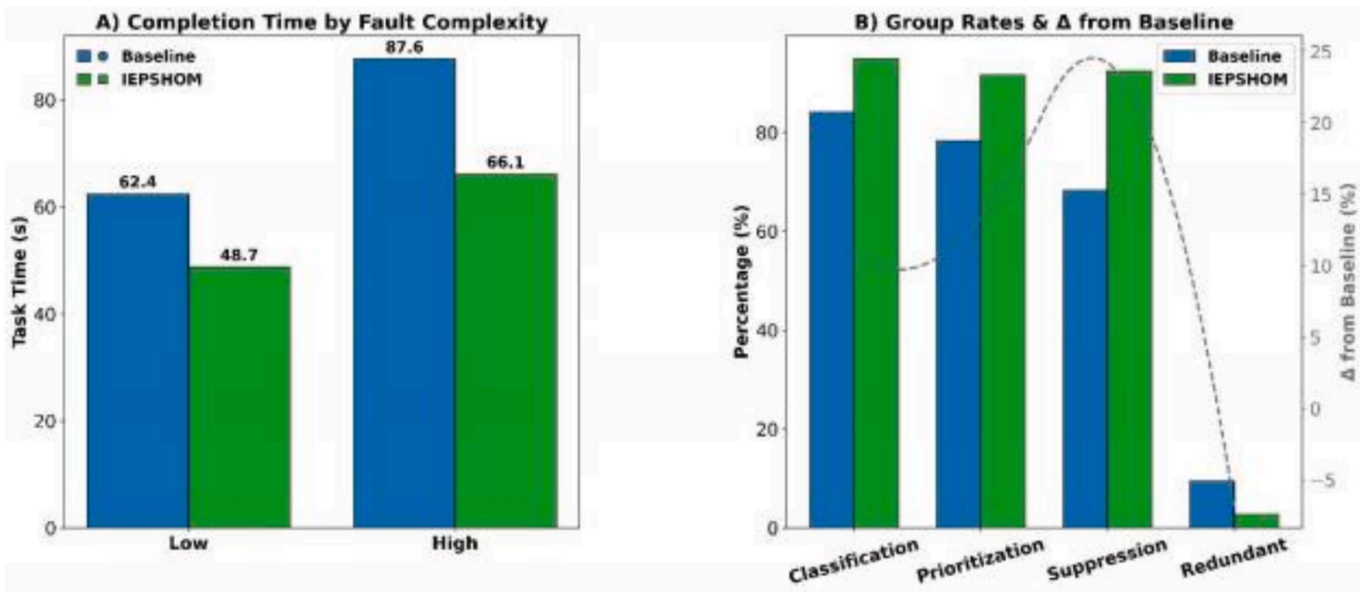

Fig. 6. A) Task completion time, missed acknowledgments, and interface switches across low- and high-complexity faults for static versus adaptive interfaces. B) Group rates for correct classification, alert prioritization, non-critical alert suppression, and redundant actions, with differences from baseline.

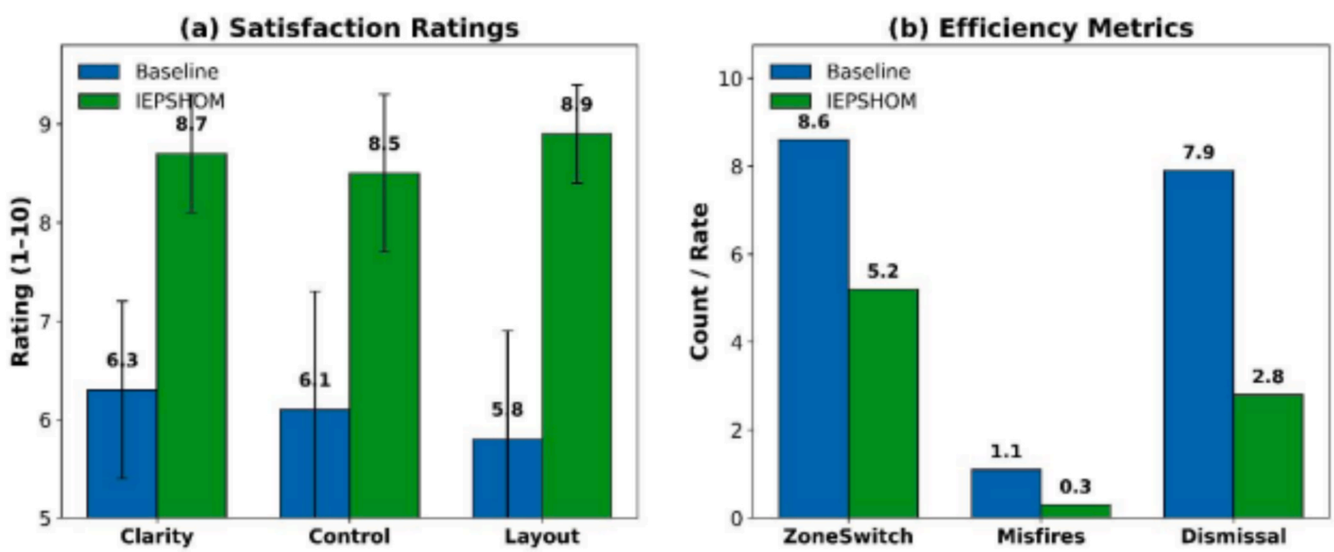

Fig. 7. A) Satisfaction ratings for clarity, control, and layout comparing Baseline and IEPSHOM with the difference trend. B) Efficiency metrics showing zone switches per task, input misfires per 10 actions, and alert dismissal errors.

The resulting classification performance (a macro-average F1-score of 97.9 and a low rate of 2.4 false-positives) is comparable to more current developments in low-latency edge circuits. Ladegourdie and Kua [19] have shown that inherently OPC-UA communication has round-trip times as low as 35 ms and stable HMI update management at 20 Hz with sub 15 percent CPU resource utilisation. In a similar instance, Pulakhandam and Nandikonda [29] reported that by placing the SCADA/HMI functions on the edge, they translated into a reduction of 60% of false alarms and the fault notification reduced to less than 200 ms, resulting in less intervention by operators. Our work is based on these results that IPC-based fault models are combined with adaptive HMI pipelines through which diagnostic alert can be rendered immediately without using queues that introduce jitter. The system avoids bus layer handoffs as it compresses the signal inference-to-visualization chain into a single hardware node, which asynchronous polling and central SCADA system process image delays has hitherto limited. Such a combination of inference and display logic at the IPC logic at the edge allows deterministic alert sequencing which is operator-trustworthy at high-load.

The major impacts of spatially appropriate HMI designs on reducing operational friction are significant zone-switching actions (−3.4 per task), input misfires (−0.8 per 10 actions), and alert misdismissal errors (−5.1%). Khan et al. [30] have discovered that the application of predictive zoning to turbine SCADA minimized operator redirection errors by 18, indicating that physical layout conformity with fault localization improves motor-cognitive targeting. Baban [31] also showed that context-based HMIs reduced manual override rates by 22 percent in general and especially in concurrent alarm conditions. These results are enhanced by our findings, which demonstrate that misfire suppression is not only a matter of operator skill but also the result of minimizing zone density and improving event-channel clarity. Inspired by the idea of eye-catching failure areas, dynamically concealing the inactive parts of the interface, and running the fault-related controls in the nearby predicted failure regions, our interface had minimized navigational ambiguity, and had suppressed the perceptual window being used to make any single decision down to a manageable level that more effectively flattened the UI entropy gradient, allowing operators to perform accurate decisions under time pressure.

The execution of collapsing inference and visualization on the edge via IPC increases performance by removing field, supervisory, and interface-to-interface layer queueing. The rule-plus-tree detector used a 100 ms loop, which enabled alert sequence determinism and minimized operator search and switching activities. Adaptive zoning further reduced misinterpretation and redundant input, suggesting that perceptual load management is a primary mediator of end-to-end response time. Larger substations and high-voltage yards introduce longer link distances, more IEDs, and higher bursty event rates, thereby increasing contention and amplifying timing drift. In such settings, IEPSHOM should be deployed with time-sensitive networking practices: VLAN and QoS separation for alarms, PTP-based time sync, OPC UA Pub/Sub (or GOOSE/SV bridging) for multicast events, and bounded ring-buffer queues sized for worst-case storm traffic. A conservative budget of ≤ 250 ms end-to-end (sensor to HMI acknowledgment) keeps operator actions within human factors guidelines while preserving our measured median of 174 ms as a target. Scaling feeders increases tag counts and concurrent inference. To avoid IPC saturation, IEPSHOM should shard computation across multiple edge nodes (one per feeder group), pin real-time threads, and offload non-urgent analytics (e.g., SHAP aggregation) to background cycles. Model execution can remain ≤ 50 ms by constraining tree depth, compiling the model, and micro-batching events during bursts; graceful degradation retains Stage-I rule checks if Stage-II classification is throttled.

The 24.2% reduction in non-critical alerts and the 5.1% decrease in alert rejection error indicate the effectiveness of local signal filtering and dynamic visual salience in the prioritization logic. Niu et al. [49] found that fallback edge processing at the time of communication reduces detection fidelity by more than 90 percent and reduces alerts by 26 percent, further confirming the importance of proximity filtering in improving relevance thresholds. Correlatively, Ferlito et al. [40] noticed that the microservice-designed HMI design that had appropriate layout responsiveness to elicit context-directed content minimized misunderstandings by an estimated 10 percent due to the design that aligned with the IEC 61724–1 standard. This is augmented by observable demand for 2.4–3.1 points of clarity and responsiveness rating improvement, to provide a more visually demoted priority of cues by involving adaptive color reweighting, blink-state temporal control, and saturation modulation. All these mechanisms reduce cognitive collisions between parallel signals and prioritize flaw-relevant information at the visual center of focus, which selects visual encoding strategies that multiply perceptual resources and trigger human attentional priority errors.

The 91.7 percent resolution of faults in under 1.5 s and the 24.9-second mean decrease in task completion time demonstrate that timing prediction and interface fluidity are vital to operational throughput. Iqbal and Baig [39] demonstrated that time-series forecasting with SCADA decreased the alert latency to less than one second, with an increase in the forecast accuracy by 14 percent, which directly resulted in faster operator responses and a reduction of HMI misfires by 15 percent. Equally, Dakheel [6] showed that sub-200 ms PLC-HMI-SCADA command cycles in a programmed laboratory environment allowed real-time command with no required manual interface input and facilitated high-frequency control procedures. Our work builds upon these premises by directly incorporating real-time task queues and sensor-fed dynamic layout recalibration into the range of events sent on the HMI. The result of this structural optimization was the decrease in indecision lag in operators and concurrent frequency of tasks hiding each other, and proved to be the critical factor when it comes to facilitating a smooth flow of cognitive handoffs between parallel control activities based on the fact that rapid convergence of interface-state behavior and not just the rate of data critically refresh matters.

At the operational level, interface-level refinements greatly facilitated the study, as the number of zone-switching actions per task decreased by 3.4, and input misfires decreased by 0.8 across 10 command attempts. Melo et al. [5] have reported steady microgrid dispatch control, in which the interactive processes between the manual interface were reduced to maintain process continuity in second-wise update loops, consistent with our aim of reducing unnecessary transitions in the most crucial tasks. Yıldırım et al. [32] showed that cognitive load in SCADA-linked edge systems might be reduced through alert clarity increments and interface simplifications, which is also reflected in our 5.1% drop in alert dismissal errors. More importantly, the correlation analysis showed that three variables of trust in the system's functionality, perception of clarity, and layout responsiveness were strongly correlated with each other ($r > 0.75$, $p < 0.01$). These results are conceptually explained by cognitive ergonomics theory, which posits that operators minimize their extraneous cognitive load and align their system suppositions with user intention. When this is incorporated into HMI zoning, cursor dynamics, and interface transition logic, the system mechanisms reduce control confusion and promote perceptual-motor coupling, thereby improving operator response fidelity under time-limited decision-state conditions.

High-voltage environments increase EMI, CT/PT saturation risk, and harmonic content, which can shift feature distributions (ΔV, θ imbalance, THD). Before rollout, IEPSHOM should (i) recalibrate thresholds with feeder-specific baselines, (ii) apply anti-alias and notch filtering tuned to site spectra, and (iii) retrain or calibrate the tree on HV datasets with domain-shift checks. Online drift monitors can trigger auto-recalibration when SHAP attribution patterns or residuals diverge from the low-voltage profile. For protection-critical yards, IEPSHOM operates as a monitoring and advisory layer and must not interfere with primary relay logic. Hot-standby IPCs, dual-NIC paths, and watchdog-based failover preserve continuity; if an edge node drops below timing guarantees, the system reverts to rule-only alerts. Configuration should align with prevailing utility standards for event time-stamping, cybersecurity (certificates, key rotation, least-privilege), and audit logging. A pragmatic path is phased scale-out: start with one bay at nominal load, then enlarge feeder groups while tracking latency percentiles, CPU headroom, and operator KPIs. Acceptance gates should include maintained F1 $\geq 97\%$ and a 90th-percentile total latency ≤ 1.2 s under synthetic fault storms.

10. Theoretical and practical Implications

This study shows that operator reactivity, system throughput, and fault diagnosis can be significantly optimized by embedding adaptive fault reasoning and perceptual interface logic within the SCADA-HMI-IPC layers of the SCADA architecture. In theory, it promotes edge-integrated automation models by recognizing that human–machine alignment is a driving force of performance. For practical purposes, it provides a deployable intelligent substation resilience framework based on decentralized decision loops and adaptive alert routing in live grid environments, supporting high load and low latency.

11. Limitations and future research

A limitation of this study is that it used a fixed hardware architecture, which limited the analysis of cross-vendor interoperability in multi-manufacturer environments. The other weakness is the fault library based on simulation, where some of the faults (rare and compound) were not considered. This will be considered in our future activities, which will include testing heterogeneous PLC and IPC vendors for semantic and time mismatches. Moreover, we will extend real-world fault capture using live utility data to evaluate generalizability across complex substation topologies.

12. Conclusion

This work validated the importance of incorporating fault detection logic, edge-hosted reasoning, and adaptive HMI interfaces into a single SCADA-IPC-PLC system architecture, thereby significantly improving real-time monitoring accuracy, reducing operational latency, and enhancing user trust and task efficiency. Across measures such as classification accuracy, operator response, and interface usability, the suggested IEPSHOM infrastructure was superior to baseline systems, especially under multi-event, high-thought, or cognitive load conditions. These results demonstrate the revolutionary potential of edge-aware, perceptually aligned control for intelligent power distribution. The study contributes to the scholarly literature by illustrating how edge-integrated architectures integrate fault analytics and adaptive HMI design into a unified framework, thereby boosting computational performance and the human–machine interface. It is credited with filling the gap between control engineering and cognitive ergonomics and with providing an evidence-based model that goes beyond technical performance to theorize on how perceptual alignment and decentralized reasoning co-evolve in complex power systems. One application in particular is regulatory requirements for more quickly containing outages, and the digital modernization of substations that focus on high-quality response performance and user interface design.

CRediT authorship contribution statement

Lingyun Gu: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fuping Wang:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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